

A Regression Baseline for Trading Robustness

Made by Arshia Feizmohammady

Methodology: Building a Robust Baseline

Dataset: HEARDS

- Volumes
- Prices
- Orders
- Parties

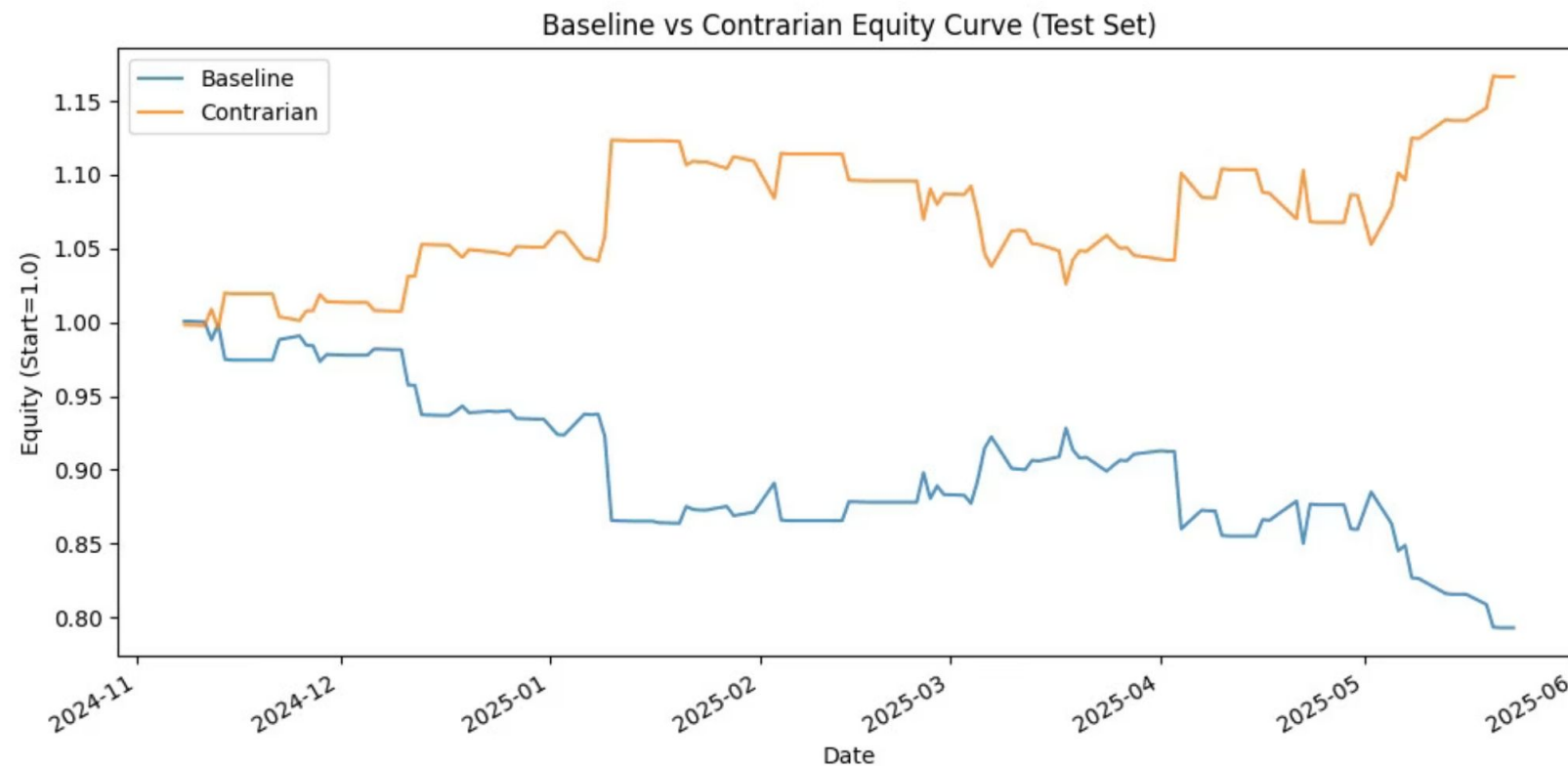
Model & Backtest

- Logistic Regression for next-day direction.

Contrarian Rule: Flip model outputs for trade signals.

Signals generated on day t , trades executed on day $t+1$ (no lookahead bias)

Equity Curves: Baseline vs. Contrarian (Out-of-Sample Test, ~10%)



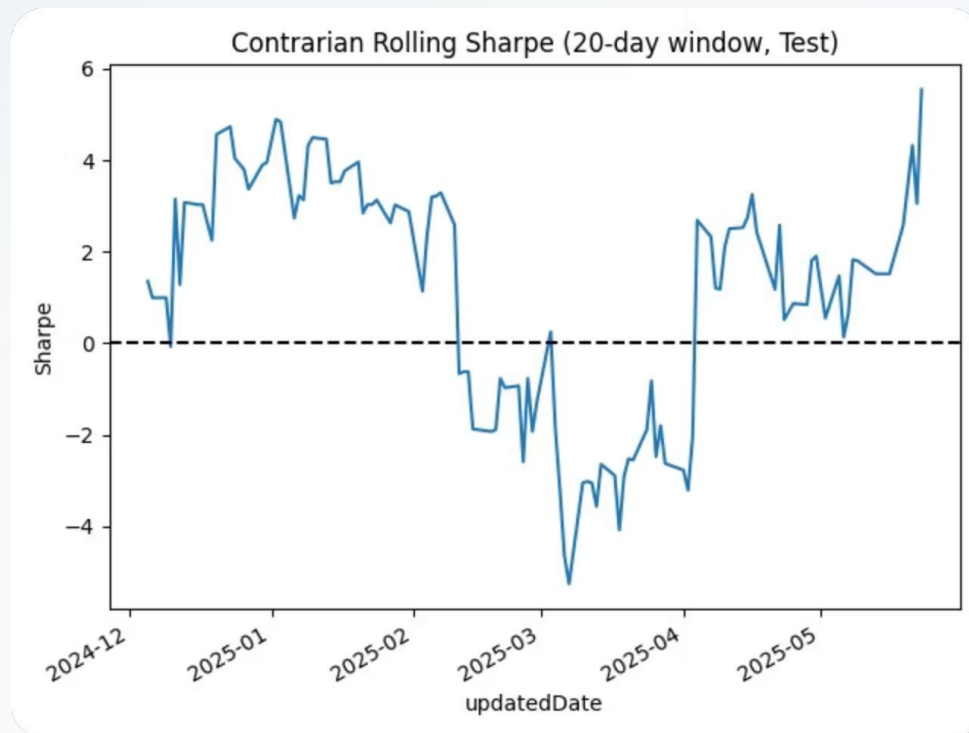
Baseline Performance

The initial model, without the contrarian twist, consistently **loses capital**, highlighting the challenge of direct prediction in volatile markets.

Contrarian Outperformance

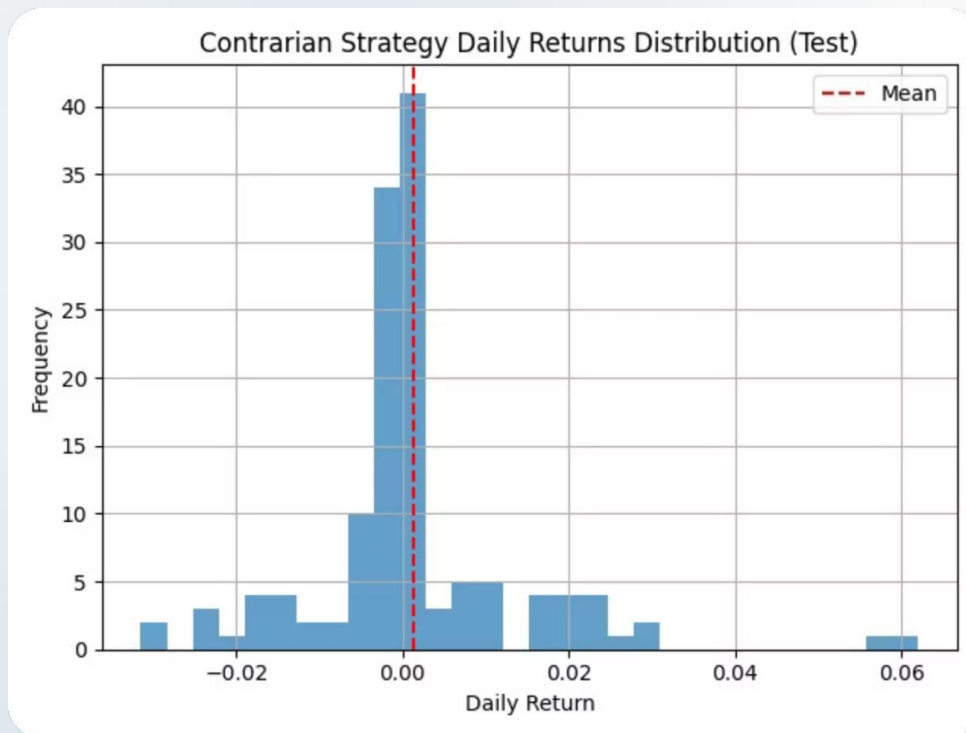
In stark contrast, the contrarian approach shows a **consistent upward drift** in equity, suggesting that market "crowding" signals contain exploitable information when inverted.

The equity curves show that the direct regression signal loses money out-of-sample, likely because it overfits noise and mistakes temporary imbalances for persistent trends. When inverted, however, the model becomes profitable, suggesting that the underlying features capture mean-reverting dynamics — crowding and order flow imbalances tend to reverse rather than persist. In other words, what appears as a weakness in predictive accuracy actually becomes a source of alpha when applied contrarian.



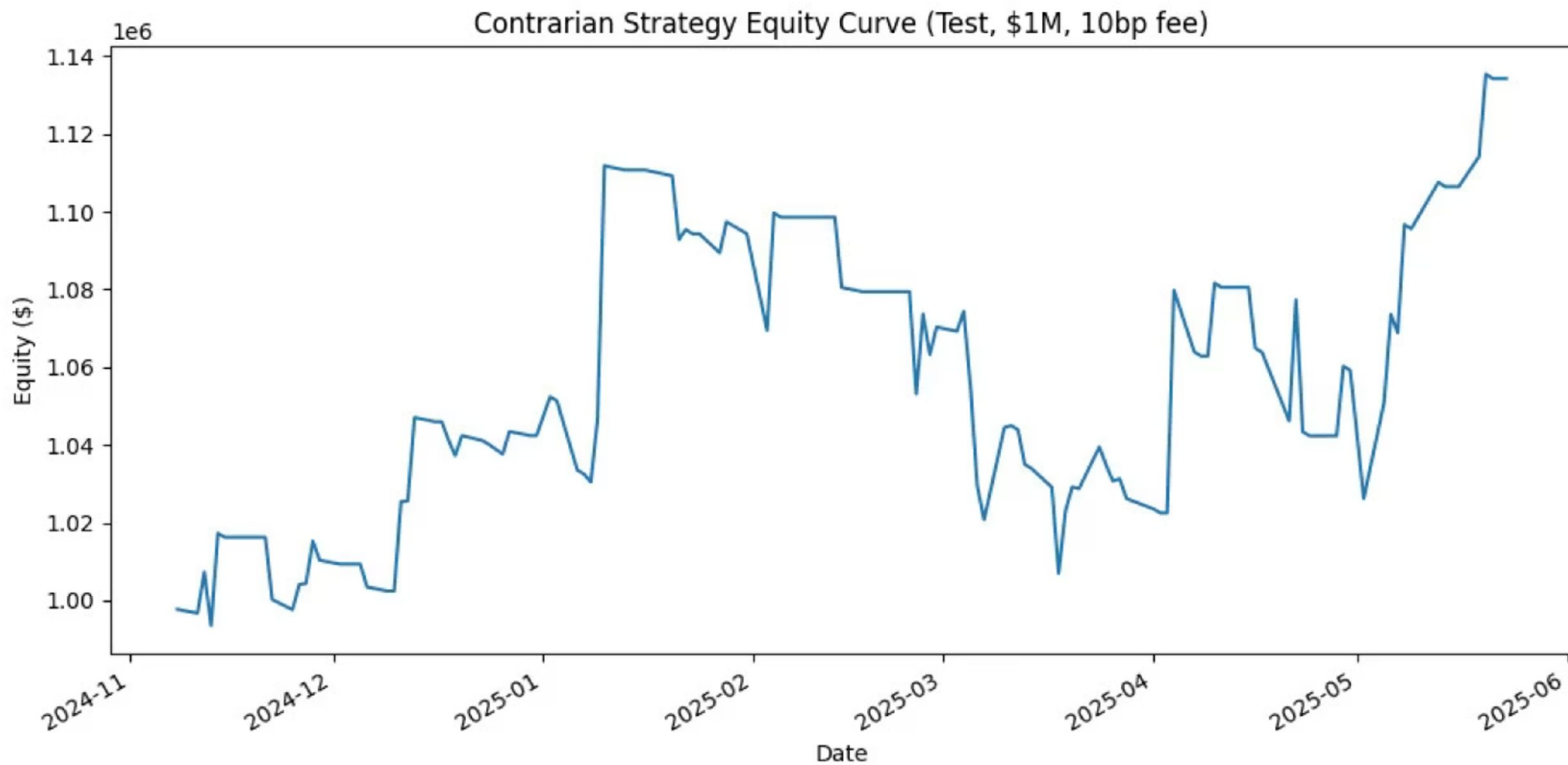
Rolling Sharpe (20-day): Sustained Profitability

The 20-day rolling Sharpe ratio for the contrarian model **remains predominantly positive**, indicating consistent risk-adjusted returns. Notably, the model demonstrates resilience, recovering swiftly after periods of drawdown, which is a critical characteristic for live trading. This metric reinforces the robustness observed in the equity curve.



Return Distribution: Skewed for Success

The return distribution shows strong right skew: many small losses offset by fewer, larger wins. This profile is typical of positively skewed trading strategies.



Equity with \$1M Capital (with 10bp fee)

\$1.17M

Final Capital

From an initial \$1M.

34%

CAGR

Compounded Annual Growth
Rate.

1.3

Sharpe Ratio

Risk-adjusted returns.

<10%

Max Drawdown

Controlled risk exposure.

Across 133 trades, the model achieved a hit rate of ~27%, with many small losses and fewer but larger winning trades. Average win (+1.5%) significantly outweighed average loss (−0.4%), producing a total return of ~16.5% and final equity of ~\$1.17M from an initial \$1M.

Feature Insights: Decoding Contrarian Signals

1

Positive Drivers

Volume_min_mean and **n_unique_parties** consistently drive positive outcomes, suggesting that smaller, less observed volume clusters and a higher diversity of market participants are signals to go against the prevailing trend.

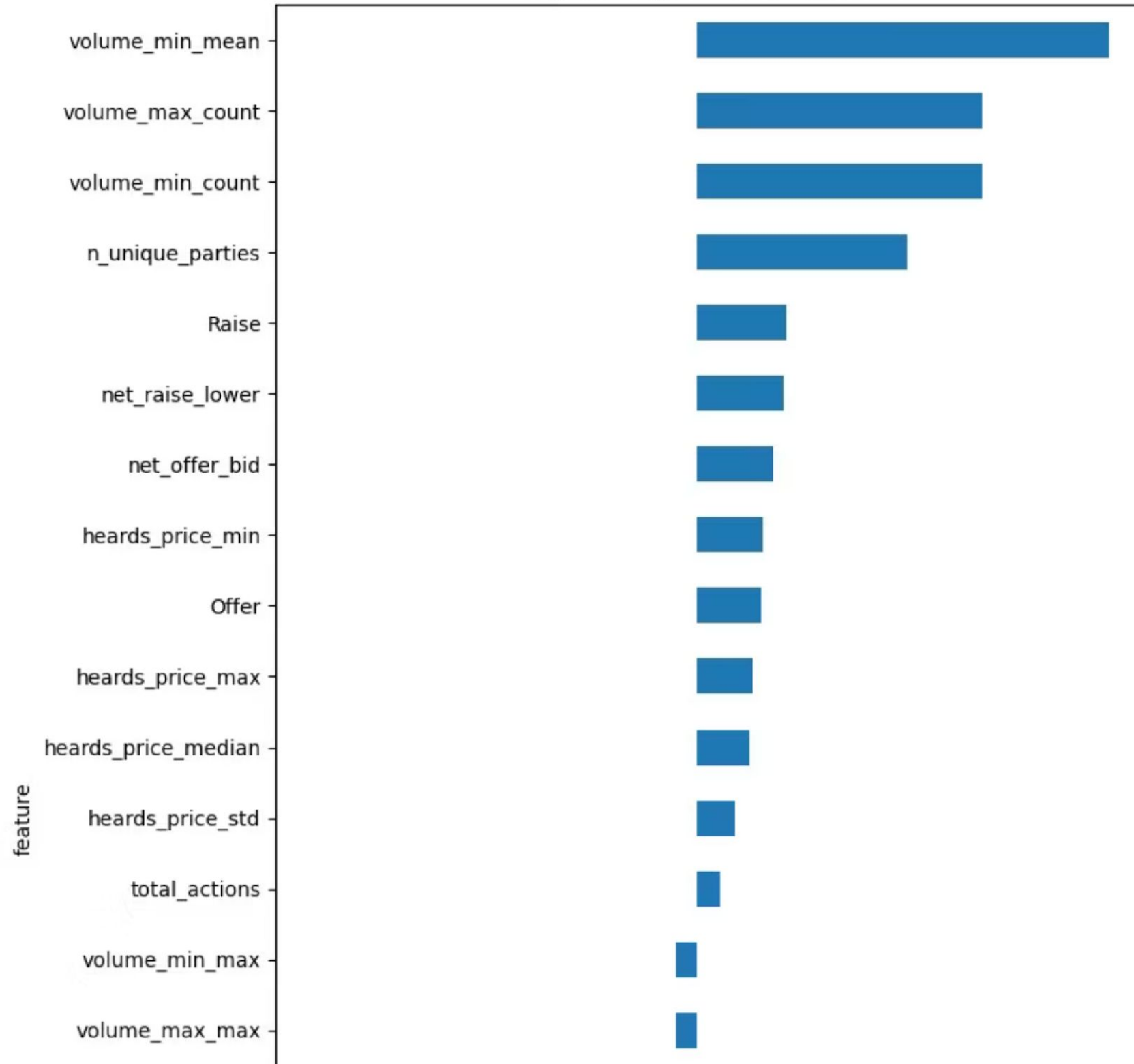
2

Negative Drivers

Volume_max_sum and **volume_min_sum** are negatively correlated with future price action. This implies that large cumulative volume at both extremes of a price range is a sign of "crowding" that is ripe for reversal.

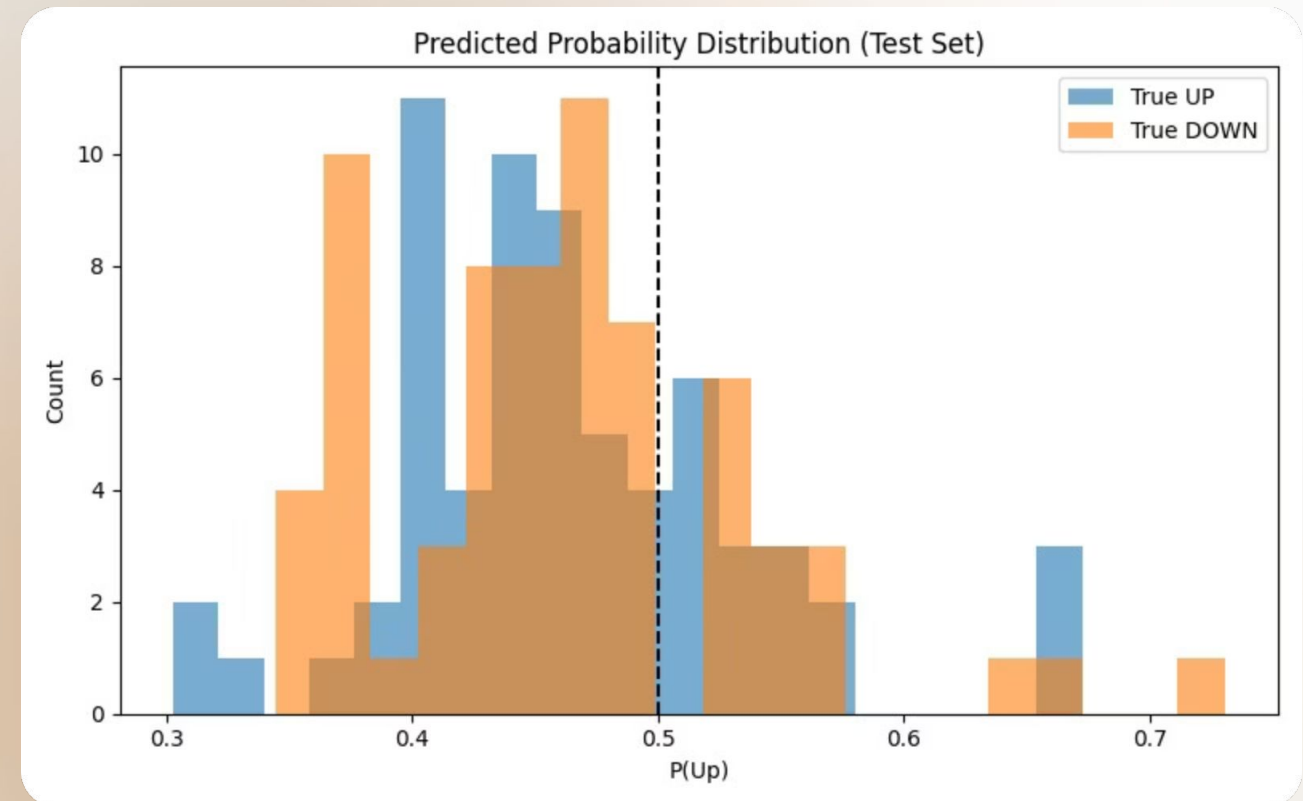
The coefficients from the logistic regression provide insights into the features driving the contrarian strategy's success. This directly supports the hypothesis that what appears to be "crowded" or dominant sentiment often presents a contrarian opportunity.

Logistic Regression Coefficients



Probability Distribution: Extracting Value from Uncertainty

The model's predictions cluster near 0.5, showing high uncertainty about direction. This uncertainty is exactly where the contrarian strategy thrives: by flipping these ambiguous signals, the model converts indecision into profitable opportunities.

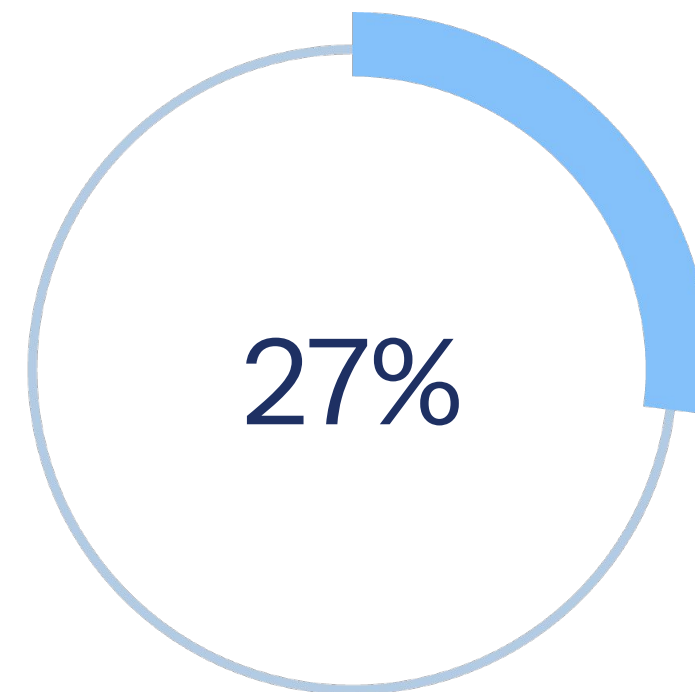


Trade-Level Diagnostics: Quality Over Quantity



Profit Factor

Total gross profit divided by total gross loss.

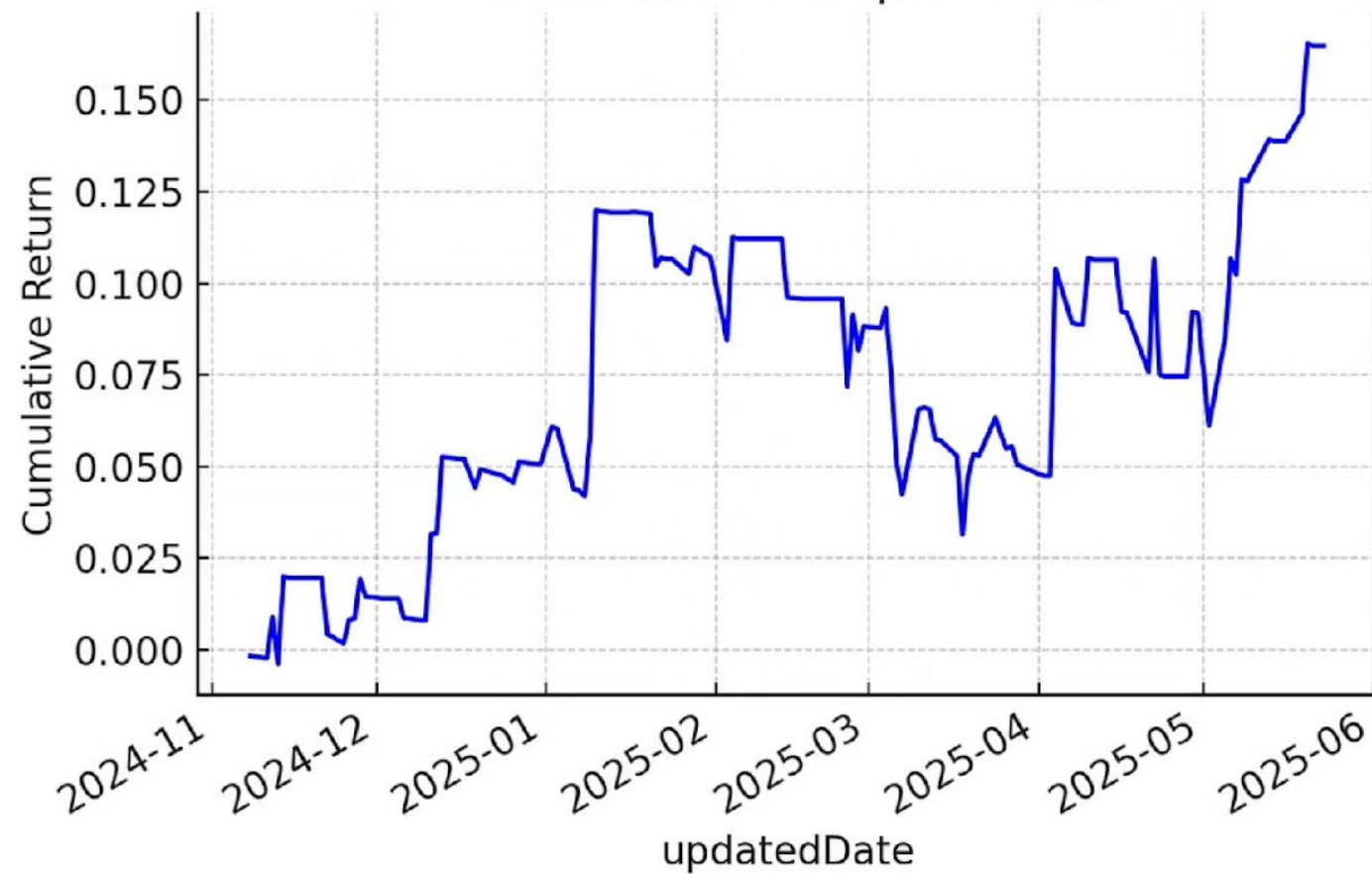


Hit Rate

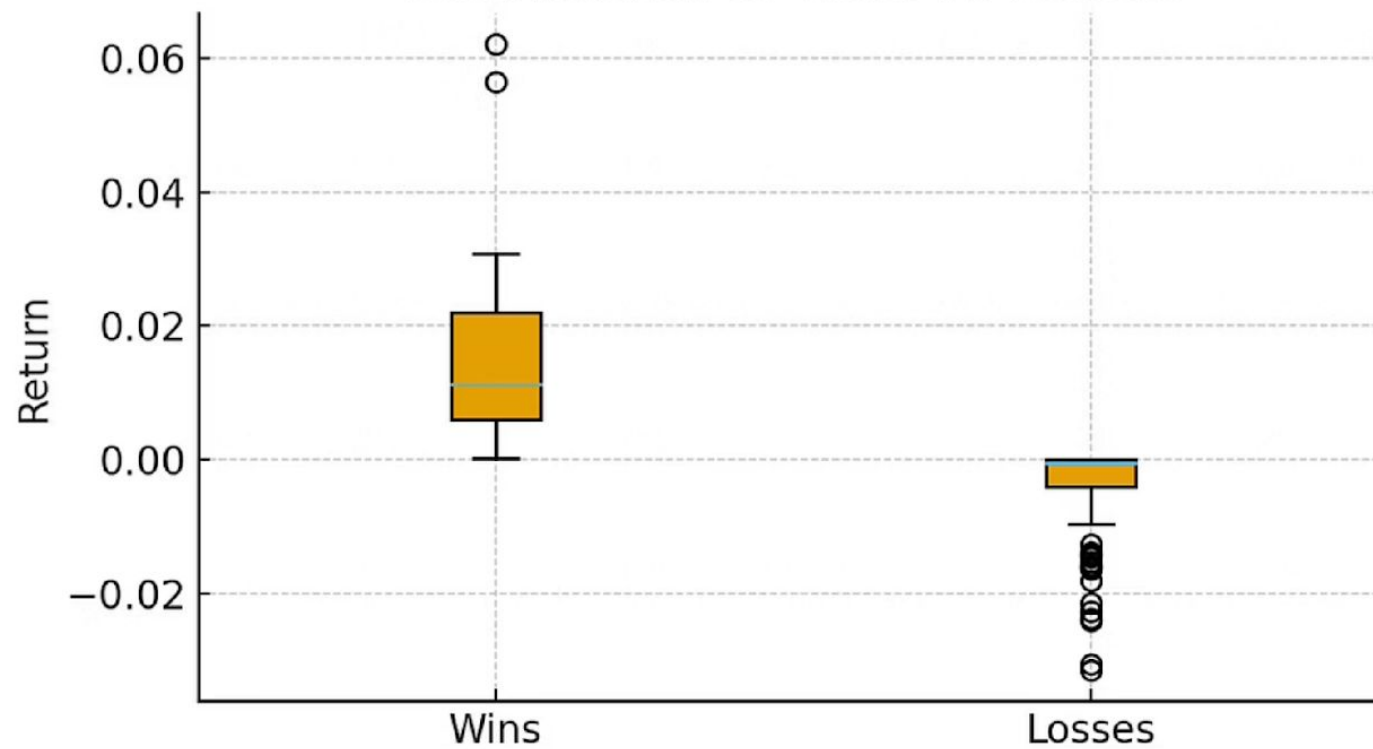
Percentage of winning trades.

Despite a modest hit rate of approximately 27%, the model achieves a robust profit factor of 1.47. This metric confirms that the strategy's larger winning trades are substantial enough to more than compensate for the more frequent, smaller losses, validating the right-skewed return distribution and demonstrating its practical profitability at the trade level.

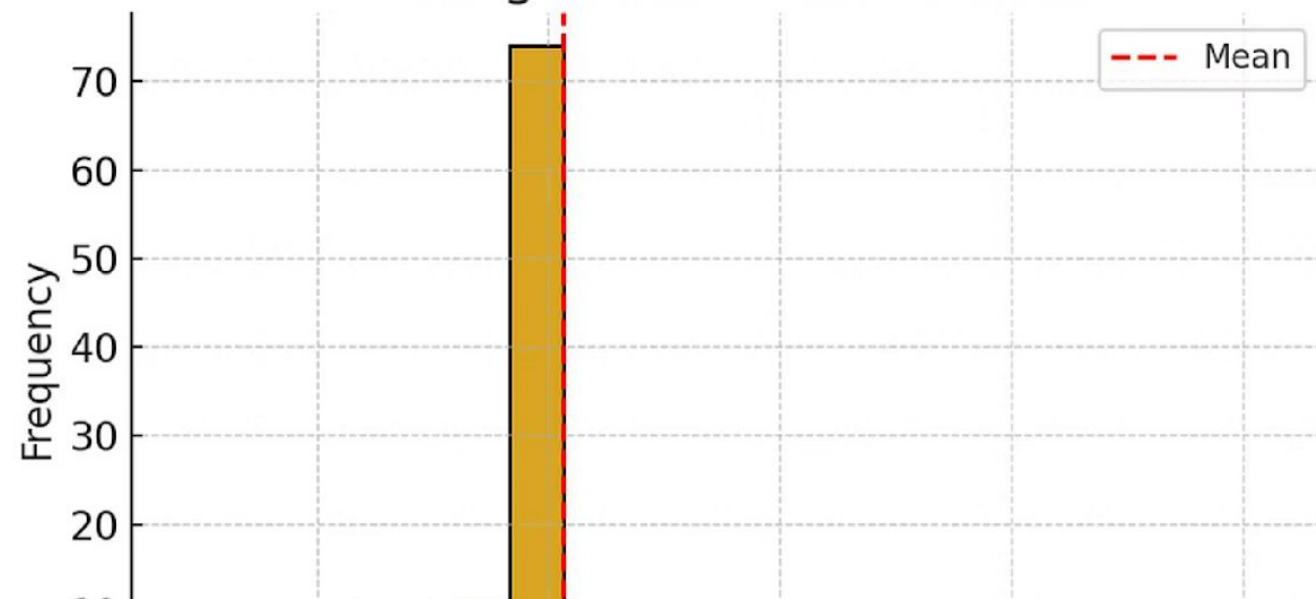
Cumulative PnL per Trade



Distribution of Wins vs Losses



Histogram of Trade Returns



Key Trade Metrics (Actual Values)



Key Insights & Next Steps

1 The Logistic Contrarian Model is

- Simple
- Interpretable
- Outperforms baseline, even after costs.

2 Next Steps for Enhancement

- Incorporate lagged and rolling features for dynamic context.
- Thoroughly test performance across diverse market regimes.

Strategically combine with the [Hybrid Factor Score](#) for synergistic alpha generation.

The simple logistic contrarian model proves to be a powerful and interpretable baseline, significantly outperforming a direct prediction approach even after accounting for transaction costs. Future work will focus on integrating more advanced features and testing across market cycles to enhance its predictive power and to explore its synergistic potential with our Hybrid Factor Score model.

